

Employee Attendance & Productivity Tracking System

1. Introduction

This project automates the weekly processing of employee attendance and productivity data using DevOps concepts and Azure Databricks.

The system processes attendance records, employee details, and task information to generate weekly analytics reports automatically.

The project uses:

- Azure Databricks
- PySpark
- Delta Live Tables (DLT)
- Unity Catalog
- YAML-based pipeline automation

2. Objective

The main objective of Week 5 is to automate the ETL workflow developed in Week 4.

The automated pipeline should:

- Run automatically every Monday
- Process attendance and task datasets
- Generate attendance metrics
- Identify top absentees
- Identify low-performing departments
- Store execution logs and reports

3. Technologies Used

Technology	Purpose
Azure Databricks	Data processing platform
PySpark	Data transformation
Delta Live Tables (DLT)	ETL orchestration
Unity Catalog	Data governance and storage
YAML	Pipeline automation configuration
GitHub	Source code management

4. Input Files

The pipeline processes three CSV files stored in Unity Catalog Volumes.

employees.csv

Contains employee master information.

Column Name	Description
employee_id	Unique employee ID
employee_name	Employee name
department	Department name
joining_date	Joining date
email	Employee email

attendance.csv

Contains employee attendance details.

Column Name	Description
attendance_id	Attendance record ID
employee_id	Employee ID
date	Attendance date
clock_in	Login time
clock_out	Logout time
status	Present or Absent

tasks.csv

Contains employee task performance details.

Column Name	Description
task_id	Task ID
employee_id	Employee ID
tasks_completed	Number of tasks completed
hours_spent	Hours spent on tasks
task_date	Task date

5. Data Storage Location

All files are uploaded into Unity Catalog Volumes.

Example path:

`/Volumes/employee_catalog/attendance_schema/raw_files/`

This provides:

- centralized storage
- secure governance
- structured data organization

6. ETL Processing Flow

The project follows Medallion Architecture using Bronze, Silver, and Gold layers.

7. Bronze Layer

The Bronze layer stores raw ingested data without transformations.

Tables:

- bronze_employees
- bronze_attendance
- bronze_tasks

Purpose:

- preserve original data
- maintain audit records
- support traceability

8. Silver Layer

The Silver layer performs data cleaning and transformation.

Transformations performed:

- remove null values
- standardize department names
- calculate employee work hours
- filter invalid records

- join attendance and task datasets
- calculate productivity score

Tables:

- silver_employees
- silver_attendance
- silver_tasks
- silver_employee_productivity

9. Gold Layer

The Gold layer generates business-ready analytics.

Tables:

- gold_department_kpis
- gold_top_performers
- gold_absentee_report

Generated metrics:

- average work hours
- department productivity
- top-performing employees
- absentee analysis

10. DevOps Automation Workflow

The DevOps workflow automates weekly ETL execution.

Workflow:

```
graph TD
    A[GitHub Repository] --> B[YAML Pipeline Trigger]
    B --> C[Weekly Scheduled Execution]
    C --> D[Databricks DLT Pipeline]
    D --> E[Gold Metrics Generation]
    E --> F[Report and Log Creation]
```

11. Weekly Scheduling

The DevOps pipeline is configured to execute automatically every Monday.

Purpose:

- automate attendance analytics
- generate weekly reports
- reduce manual execution
- maintain updated metrics

12. YAML Pipeline Configuration

Sample YAML Pipeline File

```
trigger: none
schedules:
  - cron: "0 8 * * 1"
    displayName: Weekly Monday Attendance Pipeline
    branches:
      include:
        - main
    always: true
pool:
  vmImage: ubuntu-latest
steps:
  - script: echo "Starting Attendance ETL Pipeline"
    displayName: Pipeline Start
  - script: echo "Executing Databricks DLT Pipeline"
    displayName: Run ETL
  - script: echo "Generating Attendance Reports"
    displayName: Generate Reports
  - script: echo "Pipeline Completed Successfully"
    displayName: Pipeline End
```

13. Generated Reports

The pipeline generates multiple analytics reports.

Top 5 Absentees Report

This report identifies employees with the highest absence frequency.

Example:

Employee ID	Absence Count
102	5
110	4
115	3

Purpose:

- monitor attendance issues
- identify frequent absentees

Lowest Performing Departments

This report identifies departments with low productivity scores.

Example:

Department	Avg Productivity
HR	1.8
Finance	2.2

Purpose:

- analyze department performance
- identify productivity improvement areas

Department KPI Report

This report summarizes department-level metrics.

Example:

Department	Avg Work Hours	Avg Productivity
IT	8.5	4.2
HR	7.9	2.1

Purpose:

- support management reporting
- monitor workforce efficiency

14. Log File Generation

The automated pipeline also generates execution logs.

Sample Log:

```
Pipeline Started : Monday 08:00 AM  
Bronze Layer Completed  
Silver Layer Completed  
Gold Layer Completed  
Reports Generated Successfully  
Pipeline Status : SUCCESS
```

Purpose:

- monitor execution status
- identify failures
- maintain audit history

15. Benefits of Automation

Manual Processing Challenges

- repetitive execution
- human errors
- delayed reporting
- inconsistent monitoring

Automated Pipeline Benefits

- automatic weekly execution
- faster report generation
- consistent analytics
- reduced manual effort
- better monitoring and reliability

16. Business Use Case

This project helps HR and management teams to:

- monitor employee attendance
- identify absentee trends
- evaluate department productivity

- track workforce performance
- generate weekly analytics automatically

17. Conclusion

The Week 5 DevOps implementation successfully automates the Employee Attendance ETL workflow using Azure Databricks and YAML scheduling. This automation reduces manual effort and ensures reliable weekly attendance analytics generation.

The system:

- processes attendance and task data automatically
- generates weekly reports
- identifies absentees and low-performing departments
- maintains execution logs

The project demonstrates:

- ETL automation
- DevOps workflow understanding
- YAML pipeline configuration
- Databricks DLT integration
- Medallion Architecture implementation and Business analytics reporting